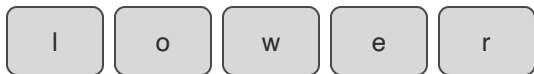


BPE Tokenization Process

Ref: GPT-2 tokenizer [Radford et al., 2019] [Sennrich et al., 2016]

Step 1: Character-level Split



Initial vocabulary: {l, o, w, e, r, ...}

□ Merge Most Frequent Pair

Step 2: Merge l+o □ lo



Vocabulary adds: {lo}

□ Continue merging

Step 3: Merge lo+w □ low, e+r □ er

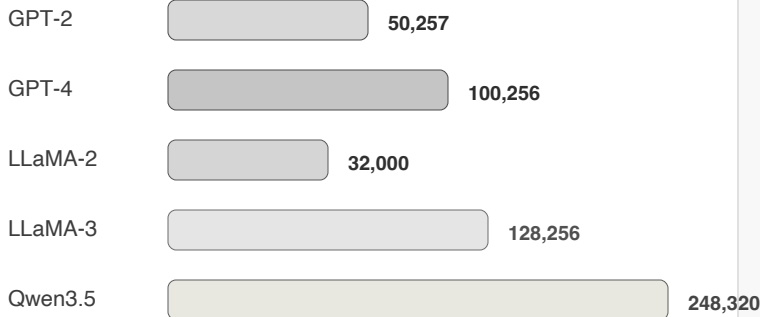


Vocabulary adds: {low, er}

"lower" □ 2 tokens: [low] [er]

Vocabulary Size Across Models

Source: Official model papers



Larger vocabulary = better Chinese encoding efficiency

GPT-2 Chinese: ~2.5 chars/token

LLaMA-3 Chinese: ~1.3 chars/token

Qwen3.5 Chinese: ~1.1 chars/token

Vocabulary size directly affects inference speed and API cost